

A1 Linear Discriminant Analysis - Answers

Linear Discriminant Analysis (LDA) - MCQs

1. What is the primary objective of Linear Discriminant Analysis (LDA)?

- ☐ A) Minimize reconstruction error
- ☐ B) Maximize class separability
- ☐ C) Minimize classification accuracy
- ☐ D) Increase dimensionality

Answer: B) Maximize class separability. LDA is supervised and seeks projections that distinguish labeled classes.

2. LDA is primarily used for:

- ☐ A) Clustering
- ☐ B) Regression
- ☐ C) Dimensionality Reduction and Classification
- ☐ D) Data Augmentation

Answer: C) Dimensionality reduction and classification. LDA can project features and also classify samples.

3. Which matrices are central to the formulation of LDA?

- ☐ A) Covariance and Correlation Matrices
- ☐ B) Within -Class Scatter and Between -Class Scatter Matrices
- ☐ C) Adjacency and Incidence Matrices
- ☐ D) Kernel and Gram Matrices

Answer: B) Within-class scatter and between-class scatter matrices.

4. The within -class scatter matrix represents:

- ☐ A) Separation between class means
- ☐ B) Variance of samples within each class
- ☐ C) Distance between clusters
- ☐ D) Prediction error

Answer: B) It measures variation of samples around their own class means.

5. The between -class scatter matrix measures:

- ☐ A) Variability within classes
- ☐ B) Separation among class means
- ☐ C) Noise variance
- ☐ D) Reconstruction loss

Answer: B) It measures separation among the class means relative to the overall mean.

6. LDA seeks projection directions that:

- ☐ A) Minimize both within -class and between -class variance
- ☐ B) Maximize within -class variance
- ☐ C) Maximize between -class variance while minimizing within -class variance

- o D) Ignore class labels

Answer: C) Maximize between-class variation while minimizing within-class variation.

7. For a dataset with C classes, the maximum number of LDA components is:

- o A) C
- o B) $C + 1$
- o C) $C - 1$
- o D) Equal to the number of features

Answer: C) $C - 1$ components at most, because the between-class scatter rank is at most $C - 1$.

8. Which of the following is an assumption of classical LDA?

- o A) Classes have identical covariance matrices
- o B) Classes have different covariance matrices
- o C) Features are independent and binary
- o D) Data must be non-linear

Answer: A) Classical LDA assumes classes share an identical covariance matrix (and commonly assumes class-conditional Gaussian distributions).

9. LDA is a supervised technique because:

- o A) It uses unlabeled data
- o B) It requires class labels
- o C) It uses reinforcement signals
- o D) It performs clustering

Answer: B) LDA uses known class labels to calculate class means and scatter.

10. In LDA, eigenvectors are obtained from:

- o A) Covariance Matrix
- o B) Correlation Matrix
- o C) $S_W^{-1} S_B$
- o D) Gram Matrix

Answer: C) The generalized eigenproblem uses $S_W^{-1} S_B$ when S_W is invertible; regularization or a generalized solver may be needed otherwise.

11. Compared to PCA, LDA:

- o A) Does not reduce dimensions
- o B) Ignores class labels
- o C) Uses class information
- o D) Requires more dimensions

Answer: C) LDA uses class information; PCA does not.

12. Which criterion is maximized in Fisher's Linear Discriminant?

- o A) Reconstruction Error
- o B) Variance Ratio
- o C) Between-Class Scatter to Within-Class Scatter Ratio

- o D) Mean Squared Error

Answer: C) The between-class scatter to within-class scatter ratio.

13. LDA assumes data from each class follows:

- o A) Uniform Distribution
- o B) Poisson Distribution
- o C) Gaussian Distribution
- o D) Exponential Distribution

Answer: C) Gaussian distributions, typically with a shared covariance matrix in classical LDA.

14. If there are two classes, the maximum number of discriminant components is:

- o A) 0
- o B) 1
- o C) 2
- o D) Number of features

Answer: B) One discriminant component at most for two classes.

15. Which statement best differentiates PCA and LDA?

- o A) PCA is supervised; LDA is unsupervised
- o B) PCA maximizes variance; LDA maximizes class separability
- o C) PCA requires labels; LDA does not
- o D) Both optimize the same objective

Answer: B) PCA maximizes overall variance; LDA seeks class separability using labels.

16. What happens if class labels are randomly shuffled before applying LDA?

- o A) Performance generally degrades
- o B) Performance improves
- o C) No effect
- o D) Number of features increases

Answer: A) Random labels generally destroy the relationship between features and classes, degrading discrimination.

17. Which matrix must generally be invertible in standard LDA?

- o A) Between -Class Scatter Matrix
- o B) Within -Class Scatter Matrix
- o C) Correlation Matrix
- o D) Distance Matrix

Answer: B) The within-class scatter matrix S_W must generally be invertible for the standard inverse formulation.

18. LDA projects data onto:

- o A) Directions of maximum variance
- o B) Random directions
- o C) Directions maximizing discrimination among classes
- o D) Kernel feature space only

Answer: C) Directions that maximize discrimination among labeled classes.

19. In pattern recognition, LDA is often applied:

- ☐ A) Before classification
- ☐ B) After classification
- ☐ C) Only during testing
- ☐ D) Only for clustering

Answer: A) LDA is commonly used before a separate classifier as supervised feature extraction.

20. Which of the following can be a limitation of LDA?

- ☐ A) Requires labeled data
- ☐ B) Assumes linear boundaries
- ☐ C) Sensitive to covariance estimation
- ☐ D) All of the above

Answer: D) All listed limitations apply: labeled data are needed, the model is linear, and covariance estimates can be unstable.

21. LDA can be viewed as:

- ☐ A) Feature Extraction Technique
- ☐ B) Data Compression Technique
- ☐ C) Classification Technique
- ☐ D) All of the above

Answer: D) LDA can be viewed as feature extraction, dimensionality reduction/data compression, and classification.

22. The rank of the between -class scatter matrix is at most:

- ☐ A) Number of Features
- ☐ B) Number of Samples
- ☐ C) Number of Classes Minus One
- ☐ D) Twice the Number of Classes

Answer: C) At most the number of classes minus one.

23. Which application commonly uses LDA?

- ☐ A) Face Recognition
- ☐ B) Speech Recognition
- ☐ C) Medical Diagnosis
- ☐ D) All of the above

Answer: D) LDA is used in face recognition, speech recognition, and medical diagnosis.

24. In Fisher's criterion, a good projection should:

- ☐ A) Bring class means closer together
- ☐ B) Increase overlap among classes
- ☐ C) Separate class means and reduce class spread
- ☐ D) Ignore variance

Answer: C) Separate class means while reducing within-class spread.

25. LDA is most effective when:

- ☐ A) Classes are well separated
- ☐ B) Data is completely random
- ☐ C) No labels are available
- ☐ D) Features are constant

Answer: A) It is most effective when class distributions are reasonably represented by its assumptions and the classes are linearly separable in the projected space.

26. Which of the following affects LDA performance?

- ☐ A) Class imbalance
- ☐ B) Outliers
- ☐ C) Small sample size
- ☐ D) All of the above

Answer: D) Class imbalance, outliers, and small sample size can all harm estimates or classification performance.

27. The generalized eigenvalue problem in LDA is solved to obtain:

- ☐ A) Cluster Centroids
- ☐ B) Principal Components
- ☐ C) Discriminant Directions
- ☐ D) Decision Trees

Answer: C) The eigenvalue problem yields discriminant directions.

28. LDA is likely to perform poorly when:

- ☐ A) Classes are linearly separable
- ☐ B) Covariance assumptions are heavily violated
- ☐ C) Labels are accurate
- ☐ D) Features are informative

Answer: B) Strong violations of covariance/distribution assumptions can make the linear discriminant suboptimal.

29. Which quantity is minimized in the denominator of Fisher's criterion?

- ☐ A) Between -Class Scatter
- ☐ B) Within -Class Scatter
- ☐ C) Total Variance
- ☐ D) Correlation

Answer: B) Within-class scatter is the denominator of Fisher's criterion.

30. In multiclass classification, LDA creates:

- ☐ A) One discriminant axis only
- ☐ B) Multiple discriminant axes up to $C-1$
- ☐ C) One axis per feature
- ☐ D) No projection axes

Answer: B) Up to $C - 1$ discriminant axes.

31. LDA is particularly useful when:

- ☐ A) Number of features is very small
- ☐ B) Feature reduction is needed before classification
- ☐ C) Data contains no labels
- ☐ D) Only clustering is required

Answer: B) LDA is useful when supervised feature reduction before classification is needed.

32. Which assumption is shared by many LDA formulations?

- ☐ A) Equal class priors
- ☐ B) Equal covariance matrices across classes
- ☐ C) Independent samples only
- ☐ D) Nonlinear decision surfaces

Answer: B) Equal covariance matrices across classes is a common assumption.

33. The output of LDA is typically:

- ☐ A) Cluster Assignments
- ☐ B) Reduced -Dimensional Feature Vectors
- ☐ C) Regression Coefficients
- ☐ D) Association Rules

Answer: B) LDA commonly returns lower-dimensional discriminant feature vectors (or predicted classes in classifier use).

34. Why is LDA often preferred over PCA for classification tasks?

- ☐ A) It ignores variance
- ☐ B) It uses label information
- ☐ C) It requires fewer samples
- ☐ D) It is unsupervised

Answer: B) It exploits labels to orient the projection toward class separation.

35. If there are 5 classes in a dataset, the maximum number of discriminant vectors produced by

LDA is:

- ☐ A) 3
- ☐ B) 4
- ☐ C) 5
- ☐ D) 6

Answer: B) With five classes, the maximum is $5 - 1 = 4$.

36. Which type of learning does LDA belong to?

- ☐ A) Supervised Learning
- ☐ B) Unsupervised Learning
- ☐ C) Reinforcement Learning
- ☐ D) Self -Supervised Learning

Answer: A) Supervised learning.

37. What role do class means play in LDA?

- ☐ A) No role
- ☐ B) Used to compute between -class scatter
- ☐ C) Used only for normalization
- ☐ D) Used only during testing

Answer: B) Class means are used to compute between-class scatter.

38. Which of the following is true about LDA decision boundaries?

- ☐ A) Always nonlinear
- ☐ B) Typically linear
- ☐ C) Always quadratic
- ☐ D) Always circular

Answer: B) Classical LDA has linear decision boundaries.

39. LDA can suffer from the "small sample size" problem when:

- ☐ A) Number of samples is much smaller than number of features
- ☐ B) Number of features is zero
- ☐ C) Classes are balanced
- ☐ D) Data is normalized

Answer: A) The small-sample-size problem occurs when the sample count is much smaller than the feature count, making scatter estimates singular or unstable.

40. Which statement about LDA is correct?

- ☐ A) It ignores class membership
- ☐ B) It maximizes variance without labels
- ☐ C) It seeks projections that improve class discrimination
- ☐ D) It is purely a clustering algorithm

Answer: C) LDA seeks projections that improve class discrimination.